

Q#4 : [4 points] Draw the circuit of 2 bit universal register

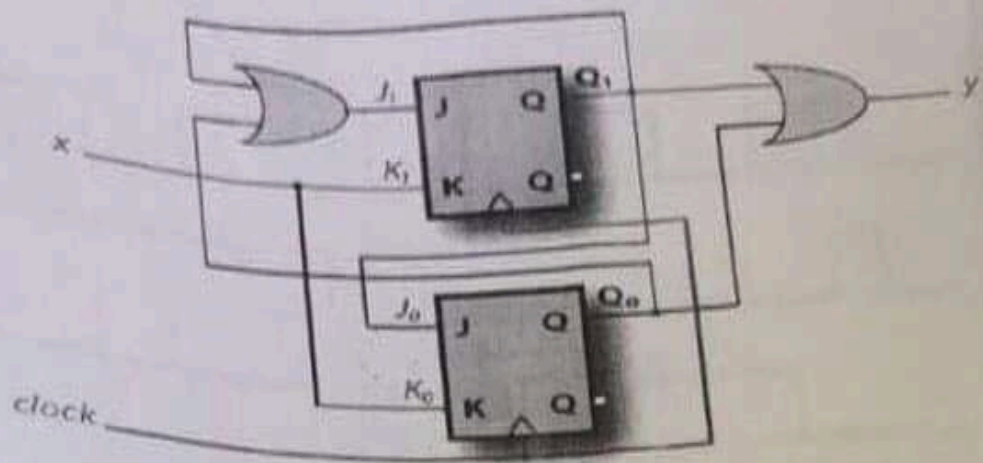
Selection line		action
S1	S0	
0	0	Shift left
0	1	Parallel load
1	0	Shift right
1	1	No change (hold data)



C.N.E
COMPUTER NETWORK
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Q#5 : [8 points]:Consider the circuit shown below

2. 2-to-4 decoders with non-inverted outputs and logic gates



1. Derive the equations

J₁=

K₁=

J₀=

K₀=

$Q_1(t+1)=Q_0(t+1)=$

2. Fill the table below

Present state		Input X	Next state				output
Q ₁	Q ₀		J ₁	K ₁	J ₀	K ₀	Y

3. Draw state diagram

Answer one question only from Q#6 and Q#7

Q #6[4 points] Draw the (Q) for the JK flip-flop with Positive Edge Triggering Clock.

Q#1 : [10.5 points] fill the table below with the

1	2	3	4	5	6	7

- The main difference between a register and a counter is _____.
 a) A register has no specific sequence of states
 b) A counter has no specific sequence of states
 c) A register has capability to store one bit of information but counter has n-bit
 d) A register counts data
- In serial shifting method, data shifting occurs _____.
 a) One bit at a time
 b) simultaneously
 c) Two bit at a time
 d) Four bit at a time
- How can parallel data be taken out of a shift register simultaneously?
 a) Use the Q output of the first FF
 b) Use the Q output of the last FF
 c) Tie all of the Q outputs together
 d) Use the Q output of all FF
- The group of bits 11001 is serially shifted (right-most bit first) into a 5-bit parallel output shift register with an initial state 01110. After three clock pulses, the register contains _____.
 a) 01110
 b) 00001
 c) 00101
 d) 00110
- Assume that a 4-bit serial in/serial out shift register is initially clear. We wish to store 1100. What will be the 4-bit in register after the second clock pulse? (left-most bit first)
 a) 1100
 b) 0011
 c) 0000
 d) 1111
- If two inputs are active on a priority encoder, which will be coded on the output?
 a) The higher value
 b) The lower value
 c) Neither of the inputs
 d) Both of the inputs
- Which gate is added to the inputs to convert OR gate into NAND gate? _____.
 a) XOR
 b) AND
 c) OR
 d) NOT

Q#2 [5 points]: Draw the state diagram of sequence detector, which detects a pattern of three or more consecutive 0's in a string of bits coming through an input line.

Q#8 : [10 points]: using T flip flops for Q2 , JK flip flop for Q1 and D flip flop for Q0 design synchronous counter that count as following 0; 1; 3; 4; 7; 0

Present state			Next state			T FF	JK FF		D FF
Q2	Q1	Q0	Q2 ⁺	Q1 ⁺	Q3 ⁺	T2	J1	K1	D0

T2=

J1=

K1=

D0=

Q#7:[4 points]: Reduce the number of states in the below state table by using row matching method

Original table

Original table
Table

Present state	Next state	
	X=1	X=0
A	A	E
B	E	F
C	G	B
D	E	F
E	D	A
F	H	G
G	G	D
H	H	G

[illegible]

7. Which gate is added to the inputs to convert OR gate into NAND gate? _____

- a) XOR b) AND c) OR d) NOT

Q#2 [5 points]: Draw the state diagram of sequence detector, which detects a pattern of three or more consecutive 0's in a string of bits coming through an input line.

Q#3 : [8.5points] Implement the equation $F(A,B,C,D) = \Sigma (0,4,5,6,7,10,11,12)$ using:

1. 4-to-1 multiplexers and logic gates.

Q.1)

1	2	3	4	5	6	7	
a	a	d	c	a 1	a	d	

Q.2)

